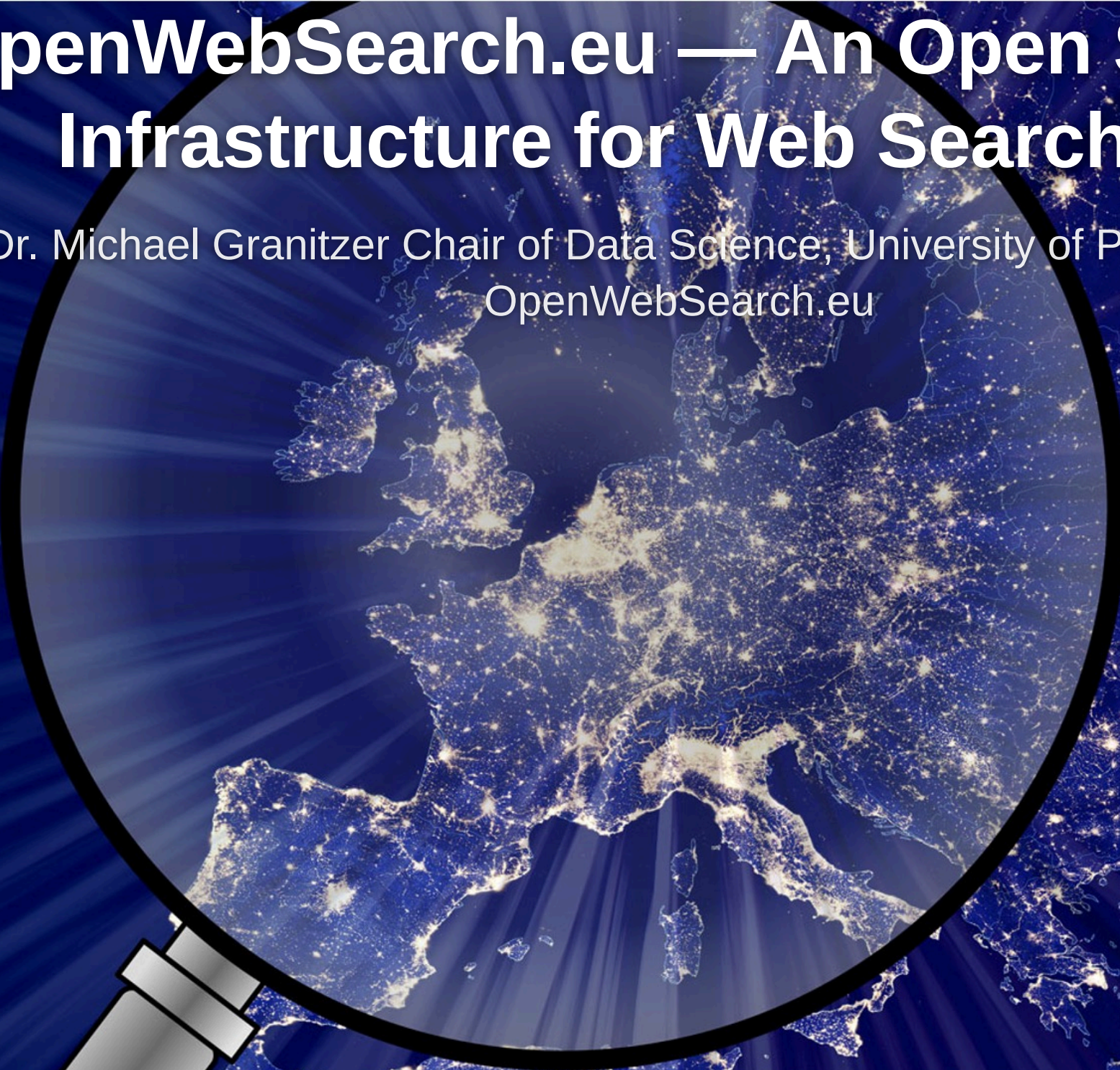


# OpenWebSearch.eu — An Open Scalable Infrastructure for Web Search & AI

Prof. Dr. Michael Granitzer Chair of Data Science, University of Passau Coordinator  
OpenWebSearch.eu



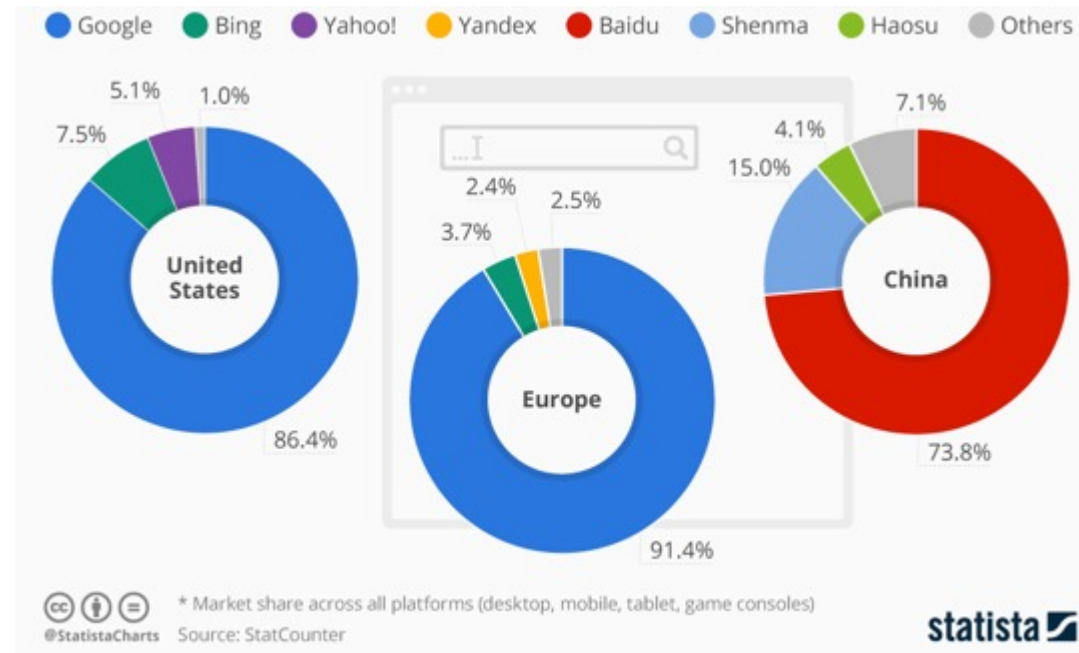


# Web Search Today

A critical resource managed as an oligopoly

## Two properties that don't fit

- A **critical infrastructure** for society, comparable to satellite navigation
- A **market oligopoly**: dominated by a small number of gatekeepers (Google, Microsoft, Baidu, ...)



# Challenges

## THE WEB: VOLUME, VELOCITY, VARIETY

Careful: domains, sites, pages, and indexed documents are different units

### VOLUME

**1.15 Billion**

Netcraft site responses

272.6 M domains · Dec 2024 Netcraft survey

**Top 0.1%–1% of sites**

may account for roughly half of all pages  
heavy-tail intuition from large web graph studies

**~ 400 Billion**

documents in Google's search index  
2020 figure cited in US v. Google testimony

*Sites ≠ domains ≠ pages; use exact counts cautiously*

### VELOCITY

**+8.6 M / month**

monthly net change in Netcraft site responses

Dec 2024 versus Nov 2024 Netcraft survey

**~ 3.2 / second**

same monthly net change, averaged across December

**149 ZB (global)**

all digital data created / copied worldwide in 2024  
IDC / Statista global datasphere estimate

*The web changes constantly; crawling is continuous, not one-off*

### VARIETY

**150+ languages**

observed across websites

W3Techs usage survey

**249 CMS Platforms**

CMS platforms identified in the HTTP Archive dataset  
HTTP Archive Web Almanac 2024 (Wappalyzer-based)

**HTML · PDF · JSON-LD · media · APIs**

content types to parse and process

*HTML · PDFs · JS apps · schemas · media · APIs*

Sources: Netcraft Web Server Survey Dec 2024 · large web graph studies (heavy-tail intuition) ·  
US v. Google testimony on ~400B documents (2020 figure) · HTTP Archive Web Almanac 2024 · W3Techs · IDC/Statista



- High infrastructure costs
- Broad range of technical skills required
  - Big Data Infrastructure
  - Natural Language Processing, Image Analysis
  - Web Technologies
- Growing legal uncertainties

# Challenges

## THE WEB: VOLUME, VELOCITY, VARIETY

Careful: domains, sites, pages, and indexed documents are different units

### VOLUME

**1.15 Billion**

Netcraft site responses

272.6 M domains · Dec 2024 Netcraft survey

**Top 0.1%–1% of sites**

may account for roughly half of all pages  
heavy-tail intuition from large web graph studies

**~ 400 Billion**

documents in Google's search index  
2020 figure cited in US v. Google testimony

*Sites ≠ domains ≠ pages; use exact counts cautiously*

### VELOCITY

**+8.6 M / month**

monthly net change in Netcraft site responses

Dec 2024 versus Nov 2024 Netcraft survey

**~ 3.2 / second**

same monthly net change, averaged across December

**149 ZB (global)**

all digital data created / copied worldwide in 2024  
IDC / Statista global datasphere estimate

*The web changes constantly; crawling is continuous, not one-off*

### VARIETY

**150+ languages**

observed across websites

W3Techs usage survey

**249 CMS Platforms**

CMS platforms identified in the HTTP Archive dataset  
HTTP Archive Web Almanac 2024 (Wappalyzer-based)

**HTML · PDF · JSON-LD · media · APIs**

content types to parse and process

*HTML · PDFs · JS apps · schemas · media · APIs*

Sources: Netcraft Web Server Survey Dec 2024 · large web graph studies (heavy-tail intuition) ·  
US v. Google testimony on ~400B documents (2020 figure) · HTTP Archive Web Almanac 2024 · W3Techs · IDC/Statista



- High infrastructure costs
- Broad range of technical skills required
  - Big Data Infrastructure
  - Natural Language Processing, Image Analysis
  - Web Technologies
- Growing legal uncertainties

High upfront costs to tap the Web as resources — especially small innovators and researchers are left behind

# But does it matter? Yes — on four dimensions

## User Choice & Democracy

- Single gatekeeper for society's information
- Search rankings can shift voting preferences by **20–72%** (SEME)
- Monopolistic ranking = single point of failure for information quality

## Untapped Richest Information Source

- 60 % of web pages carry structured data
- No open access to web-scale document collections
- Science, research and technology heavily depend on the Web as platform

## Societal & Security Risks

- European OSINT depends on US-controlled infrastructure
- Filter bubbles, algorithmic bias
- Misinformation & Disinformation

## AI & Innovation

- Web data is the fuel for LLMs, RAG, and agentic AI
- Closed indices block reproducible research & open innovation

Being able to navigate the Web is as fundamental as GPS or the power grid



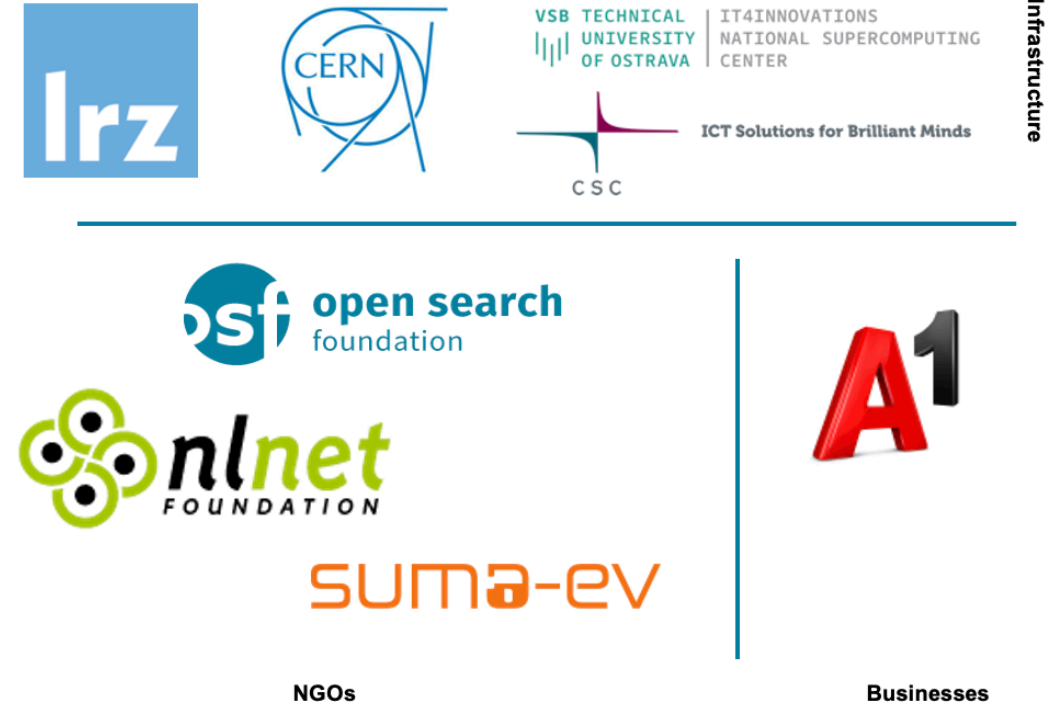
# OpenWebSearch.eu — Our Mission

Building an Open Index of the Web and an open infrastructure to reduce upfront costs for researchers, innovators and companies in tapping the Web as a resource for search, web-analytics and AI.

## Four Objectives

- 1. Open Technology Stack**  
Open-source pipelines for crawling, preprocessing, indexing
- 2. Resource Provision** Building a network of infrastructure providers (EuroHPC / AI Factories)
- 3. Added Value Services**  
Dashboard, tools, data access APIs
- 4. Bootstrapping the Ecosystem** Third-party calls, community building, applications and use-cases

## 14 Core Partners

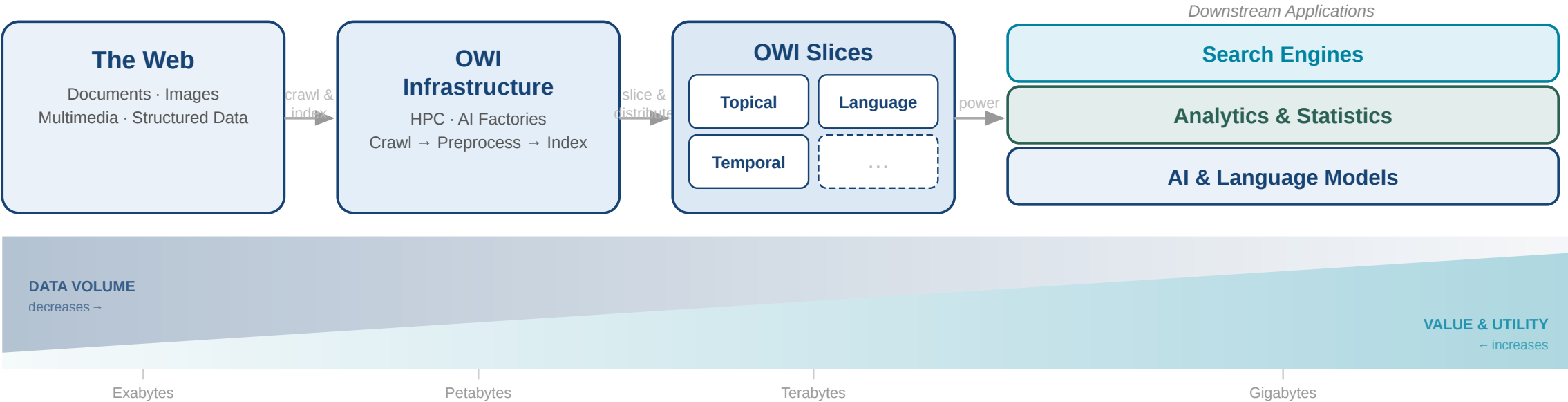


Plus 16 financially supported third-party projects (FSTPss)

# The Open Web Index (OWI)

A **Web Index** = a data structure for fast query-based access and ranking of web documents — the core of every web search engine

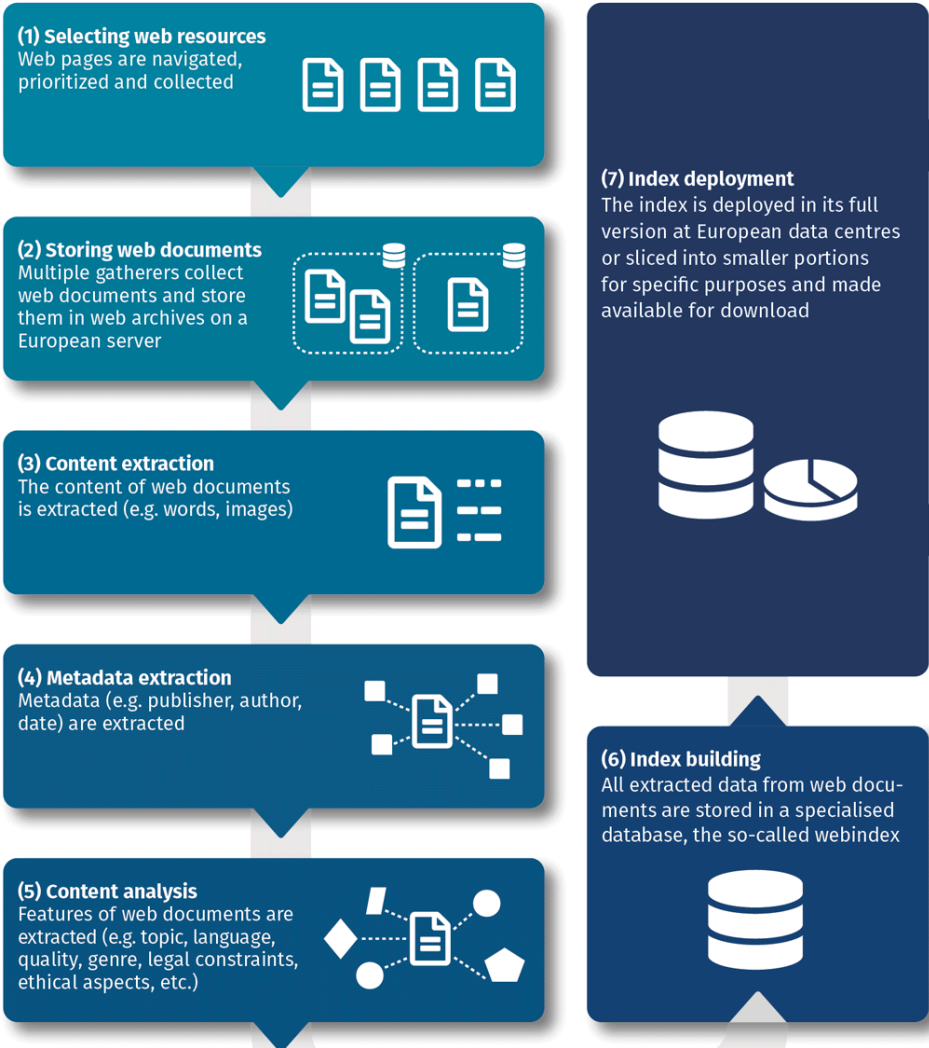
**Our Proposition:** A collaboratively created, open and transparent Web Index running on existing HPC computing centres / AI Factories in Europe



# Conceptual Workflow

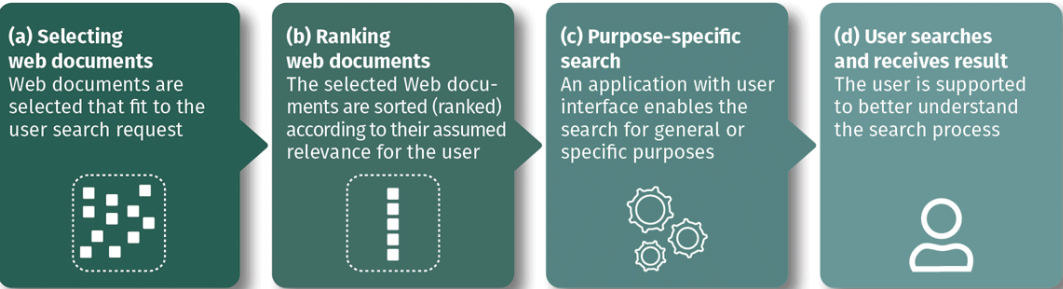
## Index Generation

Web resources are selected and retrieved, their content and metadata are analysed, and all data stored in the index database.



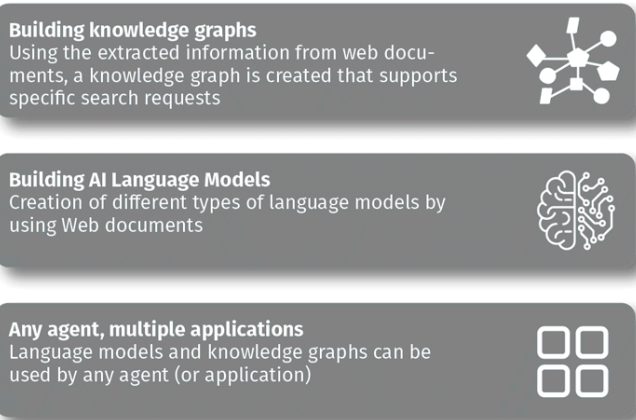
## Search Applications

A user search request will be answered by a search application that makes use of the open web index.



## Data Products

Knowledge representation models will be created using the open web index, in order to be used by any agent and for many applications

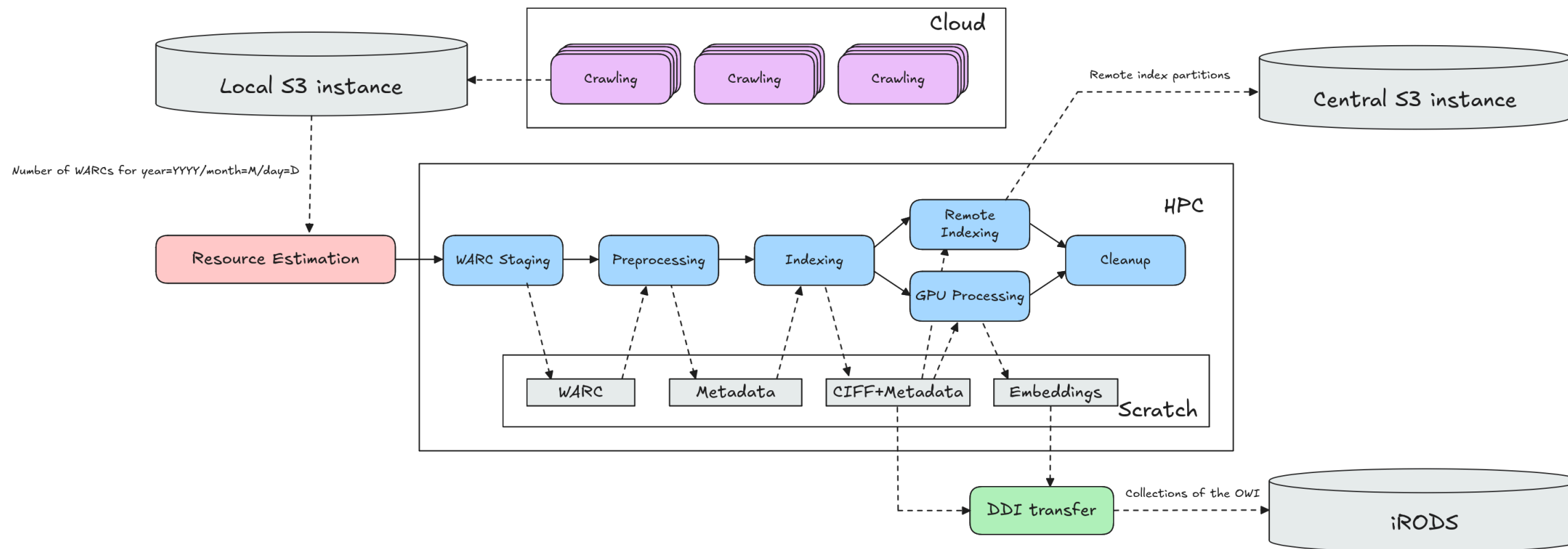


...





# Daily Workflow Pipeline



## Crawlig Infrastructure (2026)

- **25 VMs** across IT4I, LRZ, CSC, CERN
- **100 M URLs / day** crawling capacity
- **~1.1 PB** raw data · **~50 TB** index · **810+ public datasets**
- Federated storage: iRODS + S3, orchestrated via LEXIS

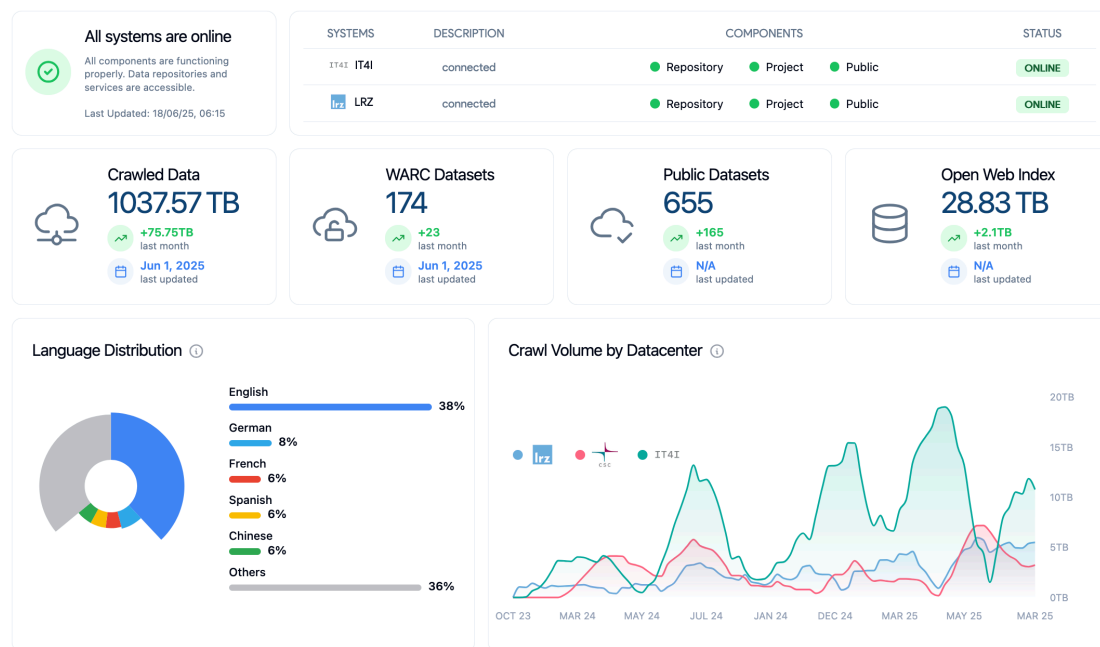
## Limitations as of 2026

- TLR 5/6: it is not a product (yet)
- No 24/7 Operations
- Limited funding - minimum engineering / improvement



# Accessing the Open Web Index

# Dashboard — Statistics & Datasets in Detail



## OWI Statistics Tab

- Repository health per data center (IT4I · LRZ)
- Key indicators: pipeline runs, HPC core-hours consumed, avg. duration, active pipelines
- Crawled data (1.3 PB), WARC datasets, public datasets, index size
- Language distribution chart + crawl volume timeline per DC
- Dataset collection distribution by resource type (main / GPU / curlie / embeddings)

## OWI Datasets Tab

- Filter: datacenter · format (WARC / Parquet / CIFF / Embeddings) · language · date range
- Infinite-scroll dataset cards: title, date, DC, public/private badge, file listing
- Parquet file preview in-browser; download via LEXIS Portal or owilix remote pull

# Dashboard — Corpora, Models & Search

## OWI Corpora

Curated, administrator-prepared collections for distribution:

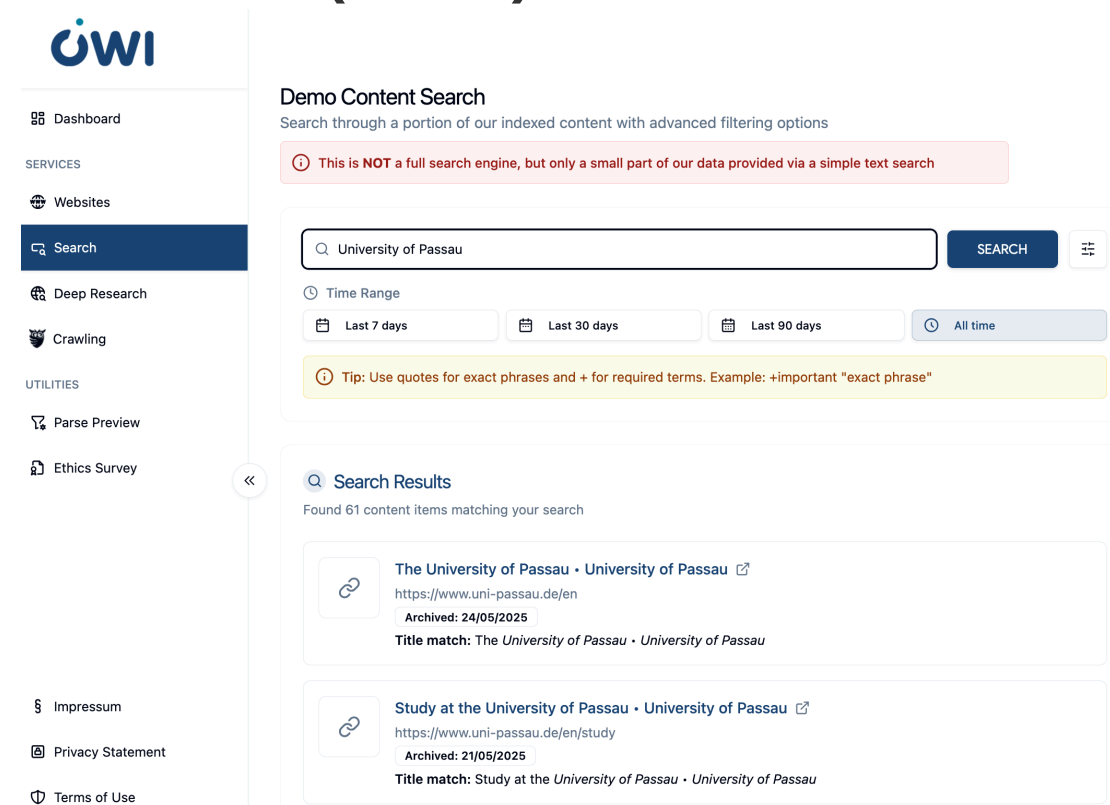
- Licensing metadata per corpus (creator, terms, full licence text)
- LEXIS integration for storage, access control and metadata
- Includes domain-specific and project-specific collections

## OWI Models

Registry of language models for web search:

- Browser-runnable models via WebAssembly (Wllama library)
- Cloud-based models (OpenAI, DeepSeek) for heavy tasks
- Model cards with training data, benchmarks,  and GGUF file listing

## OUR Search (Demo)



The screenshot shows the OWI Search Demo interface. On the left is a sidebar with the OWI logo and a navigation menu including Dashboard, Websites, Search (highlighted), Deep Research, Crawling, Parse Preview, and Ethics Survey. Below this are links for Impressum, Privacy Statement, and Terms of Use. The main content area is titled 'Demo Content Search' and includes a warning: 'This is NOT a full search engine, but only a small part of our data provided via a simple text search'. It features a search bar with 'University of Passau' entered, a 'SEARCH' button, and a 'Time Range' filter with options for 'Last 7 days', 'Last 30 days', 'Last 90 days', and 'All time'. A tip box suggests using quotes for exact phrases and '+' for required terms. The 'Search Results' section shows 'Found 61 content items matching your search' and displays two result cards. Each card includes a link icon, the title 'The University of Passau • University of Passau' and 'Study at the University of Passau • University of Passau', the URL 'https://www.uni-passau.de/en', the archive date 'Archived: 24/05/2025' and 'Archived: 21/05/2025', and a 'Title match' confirmation.

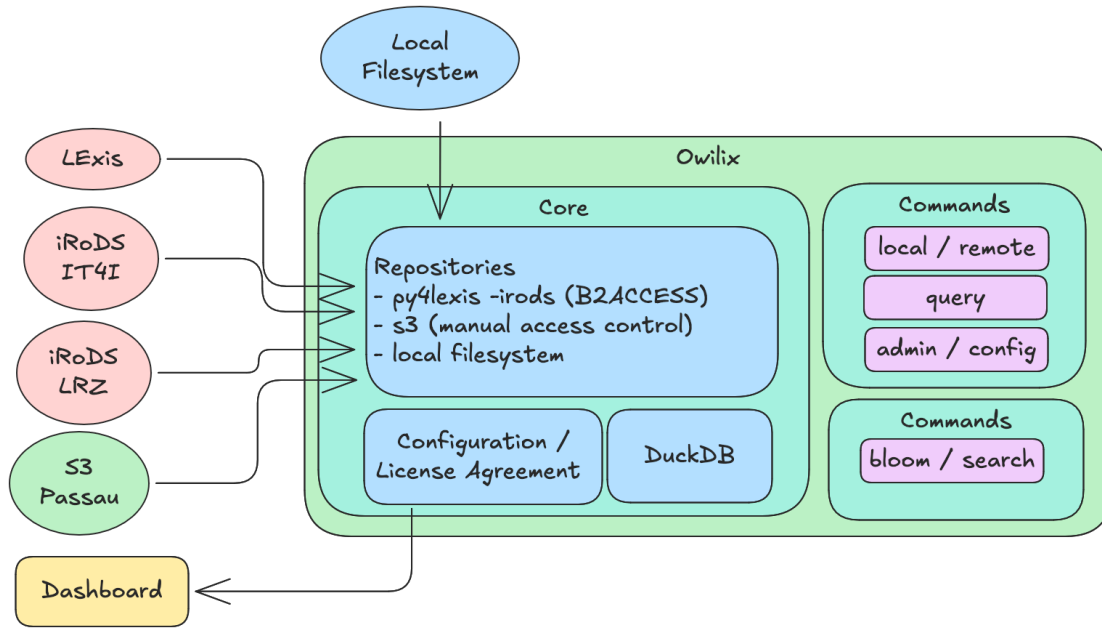
Simple + advanced field-specific search over a subset of the index; time-range filtering; result cards with thumbnail, title, URL

## Deep Research (Experimental)

AI-powered multi-step research: configurable depth + breadth, local or cloud LLM, interactive clarification mode



# owilix — The Open Web Index Client



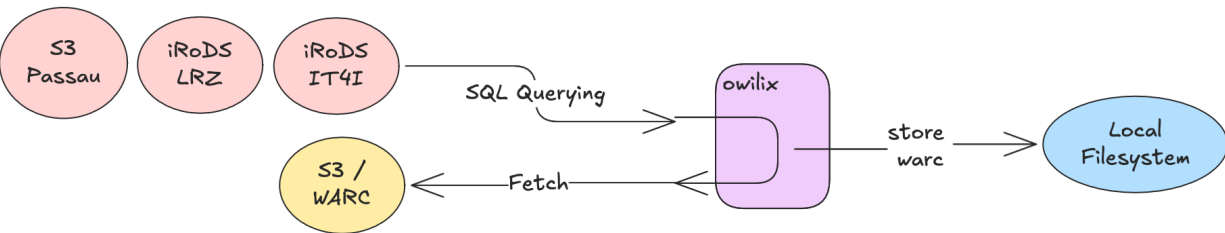
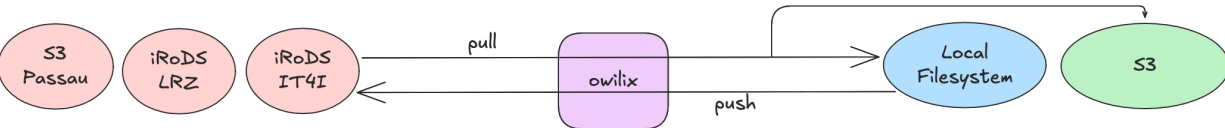
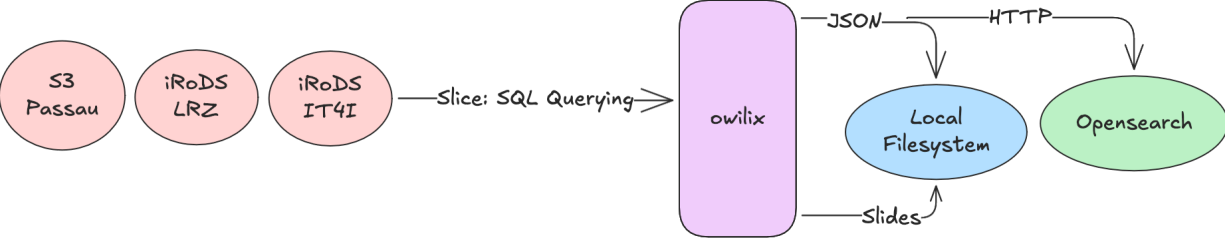
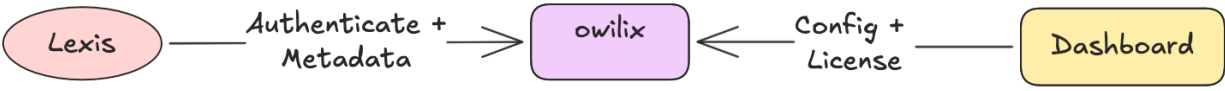
Four main workflows:

1. **Authentication** — B2ACCESS via LEXIS + Dashboard config (OpenID Connect / Keycloak)
2. **Sync Datasets** — pull/push between federated iRODS/S3 storage and local filesystem
3. **Query / Slice** — client-side SQL queries with JSON, HTTP, OpenSearch output
4. **WARC Access** — fetch original WARC files via SQL querying

Slice the index by:

- **Domain** (.de, .at, .fr, ...)
- **Topic** (Curlie.org hierarchy labels)
- **Language** (filename-based selection)
- **Sitelist** (CSV of URLs / domains / TLDs)
- **Structured data** / microdata type
- **Outlinks** for link graph analysis

# owilix Workflows



# Remote Index — Query Without Downloading

## What is the remote index?

A DuckDB / Apache Spark-queryable representation of the OWI — Parquet partitions aligned with CIFF shards, accessible from remote clients without downloading the full index.

## Example queries

```
-- Count documents per language
SELECT language, COUNT(*) as docs
FROM read_parquet('remote://lrz:latest/main/**')
GROUP BY language ORDER BY docs DESC;

-- Filter by domain
SELECT url, title FROM ... WHERE domain = 'wikipedia.org';
```

## Use cases

- Analytics and statistics on the live index
- Lightweight RAG pipeline ingestion
- Ad-hoc exploration before committing to a full  shard download

## OUR<sup>2</sup>S — Online Search API

OUR<sup>2</sup>S (built on Vespa) extends the remote index with a full REST search API:

- Structured document model: full page · snippets · embeddings
- Keyword, semantic and hybrid ranking modes
- Multi-index federation across data centers
- No local index needed — query directly over network

→ OUR<sup>2</sup>S details in the dedicated OUR<sup>2</sup>S column

## Access summary

| Mode   | Tool      | Authentication       |
|--------|-----------|----------------------|
| Browse | Dashboard | Anonymous / B2ACCESS |

| Mode                  | Tool                   | Authentication      |
|-----------------------|------------------------|---------------------|
| Download              | owilix                 | B2ACCESS<br>(LEXIS) |
| SQL queries           | DuckDB<br>remote       | B2ACCESS            |
| REST search           | OUR <sup>2</sup> S API | API key             |
| Conversational<br>RAG | MosaicRAG              | Open /<br>B2ACCESS  |

# MosaicRAG — Building RAG Systems on OWI

## From OWI Datasets to Conversational Search

MosaicRAG consumes OWI daily datasets directly — CIFF inverted index files, Parquet metadata records, and dense embeddings (published since Dec 2025) — and exposes them through a modular conversational interface.

### Pipeline for building your own MosaicRAG instance:

- 1. Download OWI slice with owilix (topic, domain, or language filter)
- 2. Ingest CIFF → inverted index; Parquet → DuckDB; embeddings → ChromaDB
- 3. Configure retrieval pipeline: sparse, dense, or hybrid
- 4. Choose reranker (TF-IDF · Embedding cross-encoder · Tournament LLM · Group LLM)
- 5. Optionally add summarization and relevance marking

Full documentation: [openwebindex.eu/book](https://openwebindex.eu/book) — "Building vertical search engines with MOSAIC"

### Supported LLMs

| Task              | Models                    |
|-------------------|---------------------------|
| Reranking         | gemma2, qwen2.5, llama3.1 |
| Summarization     | + DeepSeekv3              |
| Relevance marking | gemma2, qwen2.5, llama3.1 |

### Live Demo

Try conversational search on the Open Web Index at

[mosaicrag.ows.eu](https://mosaicrag.ows.eu)

# Using the Open Web Index — Applications

## Realised Applications

- **Mosaic & MosaicRAG** — modular vertical search framework; science, health, arts demos → [mosaic.ows.eu](https://mosaic.ows.eu)
- **Earth Observation Search** — RAG QA + geo-contextualised search (DLR / Radboud)
- **Multilingual Scientific Index** — enriching Finland's Research.fi portal (CSC)
- **Restaurant Recommender** — privacy-preserving mobile app; 214-user pilot (A1 Slovenia)
- **Argumentation Search** — args.me extended with OWI pro/contra index → [args.me/owi](https://args.me/owi)

## Community Applications (FSTPs)

- **AKSE** — argumentation knowledge graphs
- **VERITAS / DEXAI** — verification-oriented RAG
- **OMMS** — enriching OpenStreetMap with POI metadata
- **DTCommerce** — SME e-commerce product enrichment
- **CIFFIL** — cross-collection index for municipal search
- **TILDE** — trustworthy health domain search

The OWI enables a full ecosystem of vertical search engines, RAG systems, and domain-specific applications

# Using the Open Web Index — Data & Analytics

## Published Datasets

| Dataset         | Scale                 | Use case                        |
|-----------------|-----------------------|---------------------------------|
| WebFAQ          | 96M Q&A, 75 langs     | QA training, cross-lingual IR   |
| WebFAQ 2.0      | 198M pairs, 108 langs | Dense retrieval, hard negatives |
| Imprints        | 5.25M hosts, geocoded | Enterprise analytics, Eurostat  |
| German Commons  | 154B tokens           | German LLM pre-training         |
| OWS-Curlie-2025 | 20.7M docs + qrels    | IR evaluation benchmark         |

All datasets available at [openwebindex.eu/corpora](https://openwebindex.eu/corpora)

## Analytics Potential

- **Official Statistics:** enterprise data extraction from legal imprints (Eurostat, Destatis)
- **Media Monitoring:** supply chain analysis, news monitoring (JRC)
- **AI Training Data:** domain-specific corpora for LLM fine-tuning
- **Evaluation Benchmarks:** web-scale IR test collections with relevance assessments
- **Trend Analysis:** temporal web monitoring at Petabyte scale

The OWI is not just for search — it is a data infrastructure for the web-scale analytics ecosystem



# In Summary

## What we built

- A **federated, open web index** — 1.1 PB raw data, 810+ daily datasets, running 24/7 across European HPC centres
- A **full open-source pipeline** — crawl → preprocess → index → distribute, all reproducible
- A **growing ecosystem** — search apps, RAG systems, datasets, and analytics tools built on the OWI

## Why it matters

- Web data and web search are **critical infrastructure for Europe's digital sovereignty**
- The OWI **reduces upfront costs** for tapping the web as resource
- Open, transparent, and FAIR — enabling **reproducible research** and **open innovation**
- **Collaborative** approach to build and maintain the infrastructure

OpenWebSearch.eu proves that a European open web index is technically feasible, legally navigable, and economically viable

# Before we come to the Questions

# We are looking for ...

## Helping Hands

- **Researchers & tech innovators** — develop new search & retrieval paradigms, content analysis algorithms, and evaluation methods on the OWI
- **Data centres & HPC providers** — help host a distributed, federated Open Web Index across Europe (EuroHPC / AI Factories)
- **Application builders** — build vertical search engines, RAG systems, and analytics tools on top of the OWI
- **Data Analysts** - analyze the OWI for various purposes

## Helping Funds

- **Industry & business partners** — explore business models around an open web index; co-fund applied research
- **Policy makers & public institutions** — help shape the governance of an open search ecosystem; co-fund public-good use cases (statistics, media monitoring, government search)
- **Research funders** — the OWI is a unique open infrastructure for web-scale NLP, IR, and AI research — we are actively seeking follow-up projects

# Get in touch

|                 |                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------|
|                 |                                                                                                                 |
| Website         | <a href="https://openwebsearch.eu">openwebsearch.eu</a>                                                         |
| Dashboard       | <a href="https://openwebindex.eu">openwebindex.eu</a>                                                           |
| Code Repository | <a href="https://opencode.it4i.eu/openwebsearcheu-public/">https://opencode.it4i.eu/openwebsearcheu-public/</a> |
| Community       | <a href="https://openwebsearch.eu/community">openwebsearch.eu/community</a>                                     |
| Join            | <a href="mailto:join@openwebsearch.org">join@openwebsearch.org</a>                                              |
| General         | <a href="mailto:community@openwebsearch.eu">community@openwebsearch.eu</a>                                      |

Thank you — Questions, Comments, Feedback very welcome!